

Answer Key — Biology Grade 10

Sub-Strand 2.1: Plant Nutrition • Lesson 2: Leaf Structure: The Architecture of a Photosynthetic Machine

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	A	B	B	A	B	B

USE AFTER: EXPLAIN PHASE — Jigsaw expert group: palisade mesophyll function statement

Q1. In a cross-section of a mango leaf, which layer has tall cells packed closely together just beneath the upper epidermis, each holding many chloroplasts?

- A. Palisade mesophyll**
- B. Vascular bundle
- C. Lower epidermis
- D. Spongy mesophyll

Correct answer: A. Palisade mesophyll cells are tightly packed near the top of the leaf and contain many chloroplasts, so they capture the maximum amount of light for photosynthesis.

USE AFTER: OBSERVE PHASE — Step 4: locate stomata on the lower epidermis and sketch a stoma with its two guard cells

Q2. A student examines a mango leaf cross-section under a microscope and looks for stomata with guard cells. Where will she find most of them?

- A. In the lower epidermis**
- B. In the upper epidermis, where the sun is strongest
- C. Inside the palisade mesophyll layer
- D. Inside the xylem vessels

Correct answer: A. Stomata are found mainly in the lower epidermis, which is the surface the students searched in Step 4 of the observation protocol.

USE AFTER: EXPLAIN PHASE — Jigsaw expert group: spongy mesophyll / air spaces function statement

Q3. The spongy mesophyll of a leaf has many air spaces between its loosely arranged cells. What is the main function of these air spaces?

- A. They absorb sunlight for the palisade cells
- B. They let carbon dioxide diffuse from the stomata to the photosynthesising cells**
- C. They store sugars made during photosynthesis
- D. They carry water from the roots to the leaf

Correct answer: B. The connected air spaces let CO₂ entering through the stomata diffuse quickly to the mesophyll cells. They also let oxygen and water vapour move back out.

USE AFTER: EXPLAIN PHASE — Jigsaw expert group: vascular bundles function statement

Q4. A leaf vein (vascular bundle) contains xylem and phloem. Which statement correctly describes their roles?

- A. Both carry only gases in and out of the leaf
- B. Xylem brings water and minerals into the leaf, and phloem carries sugars away from the leaf**
- C. Xylem carries sugars away from the leaf, and phloem brings water and minerals in
- D. Xylem absorbs light, and phloem prevents water loss

Correct answer: B. Xylem supplies the leaf with water and minerals from the soil. Phloem exports the sugars made in photosynthesis to other parts of the plant.

USE AFTER: EXPLAIN PHASE — Synthesis question: what else moves through open stomata, and what does midday sun mean for water loss? (3-sentence written response and vote)

Q5. Kale (sukuma wiki) plants in a Nairobi garden wilt at midday even though the soil is moist, then recover the next morning. Which explanation best fits leaf structure?

- A. In strong sun the stomata open wide, so water vapour escapes faster than the xylem can supply water, and the cells lose turgor**
- B. The xylem vessels are permanently damaged by the sun and then repair themselves overnight
- C. The chloroplasts in the palisade cells burst at midday and are replaced at night
- D. The soil loses all its water at midday and gets it back at night

Correct answer: A. Open stomata let water vapour leave quickly, so water loss outpaces supply from the xylem. The cells lose turgor and the leaf wilts. When stomata close at night, water supply catches up and the plant recovers.

USE AFTER: EXPLAIN PHASE — Revisiting the sukuma wiki phenomenon: shade reduces the photosynthetic capacity of palisade cells

Q6. Kale plants moved under the shade of a mango tree stop wilting, but over several weeks the leaves turn pale yellowish-green and the plants grow slowly. Which explanation uses leaf structure correctly?

- A. Shade makes the cuticle thicker, which traps too much water in the leaf
- B. Less light reaches the palisade cells, so less photosynthesis happens, leading to pale leaves and slow growth**
- C. Shade blocks all gas exchange because the stomata are sealed shut permanently
- D. Shade makes the xylem carry sugars instead of water

Correct answer: B. Palisade cells hold most of the chloroplasts and need light to photosynthesise. In shade, photosynthesis falls, so the leaves are pale and less food is made for growth.

USE AFTER: MODEL BUILDING PHASE — Revising the Lesson 1 phenomenon model: arrows for light, water in the xylem, and water vapour through the stoma

Q7. A student revises her model of a sukuma wiki leaf by adding labelled arrows. Which set of arrows is correct?

- A. Light enters through the xylem; water moves up the cuticle; water vapour exits through the palisade cells
- B. Light enters from above through the epidermis; water moves up the xylem; water vapour exits through the stoma**
- C. Light enters from below; water moves down the xylem; sugars leave through the stoma
- D. Light enters through the lower epidermis only; water moves up the phloem; oxygen enters the xylem

Correct answer: B. Light passes in from above through the transparent cuticle and epidermis. Water travels up the xylem, and water vapour leaves through the stomata. These are the three arrows the revised model requires.